CHEMISTRY 30

UNIT A STUDY GUIDE — EXTENDED DEPTH

Thermochemical Changes

Diploma-Level: full transparency, full validation, no skipped logic.

SECTION 1 — Unit / Module Objectives

By the end of this unit, you will be able to:

- Analyze energy changes in chemical and physical processes using system vs surroundings language (q, w, ΔE), and correctly interpret exothermic vs endothermic from sign conventions.
- Derive and apply calorimetry relationships ($q = mc\Delta T$; $q = n\Delta H$; $q_{rxn} = -q_{sur}$) to determine energy changes from experimental temperature data with correct units and significant figures.
- Interpret and construct potential energy diagrams by linking activation energy, enthalpy change (ΔH), and reaction progress; justify catalyst effects on Ea without changing ΔH.
- Use Hess's Law to derive ΔH_{rxn} from a series of reactions (including phase states when given), showing algebraic manipulation of equations and enthalpy values transparently.
- Calculate ΔH_{rxn} using standard enthalpies of formation (ΔH°_f) and apply the stoichiometric summation rule: $\Delta H^\circ_{rxn} = \Sigma n\Delta H^\circ_f(\text{products}) - \Sigma n\Delta H^\circ_f(\text{reactants})$.
- Estimate reaction enthalpy using average bond enthalpies; evaluate when bond enthalpy estimates differ from formation-enthalpy results and justify why (gas-phase averages, environment dependence).
- Perform multi-step limiting-reagent thermochemistry calculations linking stoichiometry to heat (mass → moles → extent → q) with validation checks.
- Analyze and evaluate experimental design in calorimetry: identify sources of systematic/random error (heat loss, calorimeter constant, incomplete reaction) and predict how each shifts calculated ΔH.
- Interpret thermochemical data (tables, graphs, calorimeter outputs) and defend conclusions using proportional reasoning and uncertainty awareness.
- Evaluate STS applications of thermochemistry (fuels, energy efficiency, climate impacts) using structured trade-off reasoning grounded in quantitative energy evidence.

SECTION 2 — Student Self-Verification Checklist

Use this checklist to confirm diploma-level readiness:

☐ I can define system, surroundings, and universe, and I can apply sign conventions for q and ΔH correctly (exothermic: $\Delta H < 0$; endothermic: $\Delta H > 0$).

☐ I can distinguish heat vs temperature and explain why temperature change depends on heat capacity, not just heat released/absorbed.

☐ I can calculate q using $q = mc\Delta T$ and correctly convert between J and kJ; I can state and track units in every step.

☐ I can use $q_{rxn} = -q_{solution}$ (or $-q_{cal}$) and explain why the negative sign appears (energy conservation between system and surroundings).

☐ I can identify limiting reagent and connect moles reacted to energy using $q = n\Delta H$; I can justify each stoichiometric conversion.

☐ I can build a complete Hess's Law solution by reversing/scaling equations and applying the same operations to ΔH values.

☐ I can compute $\Delta H°_{rxn}$ from formation enthalpies and explicitly show product-sum minus reactant-sum with coefficients.

☐ I can estimate ΔH using bond enthalpies by calculating Σ(bonds broken) − Σ(bonds formed) and explain why the sign makes sense.

☐ I can interpret a potential energy diagram (Ea forward/reverse, ΔH, catalyst effect) and connect features to reaction behavior.

☐ I can identify diploma traps: mixing coefficients with subscripts; forgetting kJ↔J; using ΔT sign incorrectly; omitting negative sign in $q_{rxn} = -q_{sur}$; forgetting to multiply $\Delta H°_f$ by coefficients.

☐ I can validate answers: magnitude check (order of kJ), sign check (exothermic vs endothermic), and reasonableness check against physical expectations.

☐ I can write full Step 1–5 solutions with Common Error + Exam Strategy exactly as required.

SECTION 3 — Diploma-Level Practice Questions with Fully Worked Solutions

Exactly 10 questions follow. Each includes Step 1–5, Common Error, and Exam Strategy.

Question 1 | Coffee-Cup Calorimetry to Determine ΔH

Student Task:

In a coffee-cup calorimeter, 100.0 g of water initially at 22.5 °C is used to absorb heat from a reaction.

After the reaction completes, the highest stable temperature recorded is 31.8 °C.

Assume the solution behaves like water: $c = 4.184\ J\cdot g^{-1}\cdot °C^{-1}$ and density ≈ $1.00\ g\cdot mL^{-1}$.

The reaction consumed 0.0500 mol of limiting reactant.

Task: Determine ΔH for the reaction in $kJ\cdot mol^{-1}$ (sign included).

Step 1 — Identification

Given: $m_{solution} = 100.0$ g; $T_i = 22.5$ °C; $T_f = 31.8$ °C; $c = 4.184\ J\cdot g^{-1}\cdot °C^{-1}$; $n_{reacted} = 0.0500$ mol.

Find: ΔH_{rxn} ($kJ\cdot mol^{-1}$).

Relevant principle: $q_{solution} = mc\Delta T$ and energy conservation: $q_{rxn} = -q_{solution}$. Then $\Delta H = q_{rxn} / n$.

Step 2 — Principle Statement

Heat absorbed by surroundings (solution) equals heat released by the reaction (system) with opposite sign. Convert J→kJ before reporting per mole.

Step 3 — Explicit Reasoning

Compute $\Delta T = T_f - T_i = 31.8 - 22.5 = +9.3$ °C.

Compute $q_{solution} = mc\Delta T = (100.0\ g)(4.184\ J\cdot g^{-1}\cdot °C^{-1})(9.3\ °C) = 3891$ J.

Convert: $q_{solution} = 3.891$ kJ (since 1000 J = 1 kJ).

Apply sign: $q_{rxn} = -q_{solution} = -3.891$ kJ (reaction released heat because solution warmed).

Compute $\Delta H_{rxn} = q_{rxn} / n = (-3.891\ kJ) / (0.0500\ mol) = -77.8\ kJ\cdot mol^{-1}$.

Pause and predict: Because temperature increased, the reaction must be exothermic → ΔH should be negative. This matches our sign.

Step 4 — Validation

Sign check: temperature rose → exothermic → $\Delta H < 0$ (passes).

Magnitude check: tens to hundreds of $kJ\cdot mol^{-1}$ is typical for many reactions (passes).

Unit check: kJ divided by mol gives $kJ\cdot mol^{-1}$ (passes).

Step 5 — Final Answer

$\Delta H_{rxn} = -77.8\ kJ\cdot mol^{-1}$.

Common Error

Using $\Delta T = T_i - T_f$ (wrong sign) and then also applying the negative sign in $q_{rxn} = -q_{solution}$, causing a double sign error.

Leaving q in joules but writing $kJ \cdot mol^{-1}$ in the final answer (unit mismatch).

Exam Strategy

Write a sign sentence before calculating: “If T increases, reaction is exothermic, so ΔH must be negative.” Then check your final sign against it.

Use a unit roadmap: g → J → kJ, then kJ ÷ mol → $kJ \cdot mol^{-1}$.

Question 2 | Using a Calorimeter Constant

Student Task:

A metal calorimeter has a calorimeter constant $C_{cal} = 220\ J \cdot °C^{-1}$.

A reaction occurs inside the calorimeter and the temperature of the calorimeter rises from 19.60 °C to 26.15 °C.

The reaction consumed 0.125 mol of limiting reactant.

Task: Determine ΔH_{rxn} in $kJ \cdot mol^{-1}$ (include sign).

Step 1 — Identification

Given: $C_{cal} = 220\ J \cdot °C^{-1}$; $T_i = 19.60$ °C; $T_f = 26.15$ °C; n = 0.125 mol.

Find: ΔH_{rxn} ($kJ \cdot mol^{-1}$).

Relevant principle: $q_{cal} = C_{cal}\Delta T$ and $q_{rxn} = -q_{cal}$.

Step 2 — Principle Statement

For a calibrated calorimeter, heat absorbed by calorimeter equals $C_{cal}\Delta T$. The reaction releases/absorbs the opposite amount.

Step 3 — Explicit Reasoning

$\Delta T = 26.15 - 19.60 = +6.55$ °C.

$q_{cal} = (220\ J\cdot°C^{-1})(6.55\ °C) = 1441\ J = 1.441\ kJ$.

$q_{rxn} = -1.441$ kJ (temperature increased → exothermic).

$\Delta H_{rxn} = q_{rxn} / n = (-1.441\ kJ)/(0.125\ mol) = -11.5\ kJ\cdot mol^{-1}$ (3 sig figs).

Step 4 — Validation

Sign check: calorimeter warmed → reaction exothermic → negative ΔH (passes).

Unit check: J → kJ and kJ ÷ mol → $kJ\cdot mol^{-1}$ (passes).

Step 5 — Final Answer

$\Delta H_{rxn} = -11.5\ kJ\cdot mol^{-1}$.

Common Error

Using $q = mc\Delta T$ even though the problem gives a calorimeter constant (wrong model).

Forgetting to convert J to kJ.

Exam Strategy

Circle whether you are given mass+specific heat OR calorimeter constant. Use only the model the problem provides.

Always compute ΔT explicitly with sign before doing q.

Question 3 | Limiting Reagent + Heat Released

Student Task:

Hydrogen reacts with oxygen to form liquid water: $2H_2(g) + O_2(g) \rightarrow 2H_2O(l)$.

The enthalpy change for the reaction as written is $\Delta H = -572$ kJ.

If 5.00 g of H_2 reacts with 40.0 g of O_2, determine the total heat released (kJ).

Assume complete reaction and identify the limiting reagent.

Step 1 — Identification

Given: 5.00 g H_2 and 40.0 g O_2; reaction: $2H_2 + O_2 \rightarrow 2H_2O$ with $\Delta H = -572$ kJ per 2 mol H_2 (reaction as written).

Find: total heat released (q_{rxn}) for the limiting amount reacted.

Relevant principles: limiting reagent via mole comparison; then scale ΔH by reaction extent.

Step 2 — Principle Statement

Convert masses to moles, use stoichiometric coefficients to find which reactant limits the reaction extent. Heat released scales proportionally with the number of 'reaction moles' completed.

Step 3 — Explicit Reasoning

1) Convert to moles: $n(H_2) = 5.00\ g \div 2.02\ g\cdot mol^{-1} = 2.48$ mol.

$n(O_2) = 40.0\ g \div 32.00\ g\cdot mol^{-1} = 1.25$ mol.

2) Compare to coefficients (2 mol H_2 required per 1 mol O_2).

For 1.25 mol O_2, required $H_2 = 2(1.25)=2.50$ mol H_2.

Available $H_2 = 2.48$ mol < 2.50 mol $\rightarrow$ H_2 is limiting.

3) Determine reaction extent: the reaction as written consumes 2 mol H_2 per -572 kJ.

Extent factor $= n(H_2)/2 = 2.48/2 = 1.24$ 'reaction units'.

4) Heat released: $q_{rxn} = (1.24)(-572\ kJ) = -709$ kJ.

Pause and predict: since the reaction forms water and is known to be strongly exothermic, q should be negative and large in magnitude. That matches.

Step 4 — Validation

Limiting check: If H_2 is limiting, some O_2 remains; required O_2 for 2.48 mol H_2 is 1.24 mol O_2, which is less than available 1.25 mol (consistent).

Sign check: exothermic → negative q (passes).

Step 5 — Final Answer

Total heat released: 709 kJ released ($q_{rxn} = -709$ kJ).

Common Error

Scaling ΔH by total moles of reactant without accounting for coefficients (e.g., using 2.48 mol directly with −572 kJ).

Choosing limiting reagent by mass instead of moles/coefficients.

Exam Strategy

Always do limiting with moles and coefficients: compute how much of the other reactant is required. Then scale ΔH by 'reaction units' (moles ÷ coefficient).

Question 4 | Hess's Law to Find ΔH of CO Combustion

Student Task:

Use the following thermochemical equations to determine ΔH for the combustion of carbon monoxide:

(1) $C(s) + O_2(g) \rightarrow CO_2(g)$ $\Delta H = -394$ kJ

(2) $C(s) + 1/2\ O_2(g) \rightarrow CO(g)$ $\Delta H = -111$ kJ

Target: $CO(g) + 1/2\ O_2(g) \rightarrow CO_2(g)$ $\Delta H = ?$

Show all algebraic steps clearly.

Step 1 — Identification

Given two equations with ΔH values. Need ΔH for $CO + 1/2\ O_2 \rightarrow CO_2$.

Relevant principle: Hess's Law — if you add reactions, you add enthalpy changes; if you reverse a reaction, you change the sign of ΔH; if you multiply coefficients, multiply ΔH.

Step 2 — Principle Statement

Construct the target equation by algebraically combining the provided equations. Apply the same operations to ΔH.

Step 3 — Explicit Reasoning

Target has CO on the reactant side and CO_2 on the product side. Equation (2) produces CO; reverse it to place CO as a reactant:

Reversed (2): $CO(g) \rightarrow C(s) + 1/2\ O_2(g)$ $\Delta H = +111$ kJ.

Now add reversed (2) to equation (1):

(1) $C(s) + O_2(g) \rightarrow CO_2(g)$ $\Delta H = -394$ kJ

(2R) $CO(g) \rightarrow C(s) + 1/2\ O_2(g)$ $\Delta H = +111$ kJ

Add and cancel species that appear on both sides (C(s) cancels):

$CO(g) + O_2(g) \rightarrow CO_2(g) + 1/2\ O_2(g)$

Subtract $1/2\ O_2(g)$ from both sides to match target:

$CO(g) + 1/2\ O_2(g) \rightarrow CO_2(g)$.

Add enthalpies: $\Delta H = (-394) + (+111) = -283$ kJ.

Step 4 — Validation

Check that the target equation is correct and balanced: C and O atoms conserved (passes).

Sign check: CO combustion is exothermic → negative ΔH (passes).

Step 5 — Final Answer

$\Delta H = -283$ kJ for $CO(g) + 1/2\ O_2(g) \rightarrow CO_2(g)$.

Common Error

Forgetting to reverse equation (2) before adding, which puts CO on the wrong side.

Manipulating the equation but not applying the same sign change to ΔH.

Exam Strategy

Write the target equation at the top. Then force each given equation to ‘move’ substances to the correct side by reversing first, scaling second, and only then adding.

Question 5 | ΔH° from Enthalpies of Formation

Student Task:

Use standard enthalpies of formation to calculate ΔH°_rxn for:

$CH_4(g) + 2O_2(g) \rightarrow CO_2(g) + 2H_2O(l)$

Given: ΔH°_f [$CH_4(g)$] = −75.0 kJ·mol−1; ΔH°_f [$CO_2(g)$] = −394 kJ·mol−1; ΔH°_f [$H_2O(l)$] = −286 kJ·mol−1; ΔH°_f [$O_2(g)$] = 0.

Show the full summation with coefficients.

Step 1 — Identification

Given formation enthalpies for species. Need ΔH°_rxn for combustion of methane.

Relevant principle: ΔH°_rxn = ΣnΔH°_f(products) − ΣnΔH°_f(reactants). Elements in standard state have ΔH°_f = 0.

Step 2 — Principle Statement

Multiply each ΔH°_f by its stoichiometric coefficient, sum products, sum reactants, then subtract.

Step 3 — Explicit Reasoning

Products sum: [CO_2] + 2[$H_2O(l)$] = (1)(−394) + (2)(−286) = −394 −572 = −966 kJ.

Reactants sum: [CH_4] + 2[O_2] = (1)(−75.0) + (2)(0) = −75.0 kJ.

$\Delta H°_{rxn} = (-966) - (-75.0) = -891$ kJ.

Pause and predict: Combustion is strongly exothermic, so ΔH should be negative with large magnitude; −891 kJ is reasonable.

Step 4 — Validation

Check coefficients were applied to $\Delta H°_f$ (yes). Check O_2 is standard state so $\Delta H°_f=0$ (yes). Sign check exothermic (passes).

Step 5 — Final Answer

$\Delta H°_{rxn} = -891$ kJ (per 1 mol CH_4 combusted as written).

Common Error

Forgetting to multiply $\Delta H°_f$ of water by 2, giving a value that is too small in magnitude.

Subtracting in the wrong direction (reactants − products).

Exam Strategy

Write two labeled lines: Σproducts and Σreactants. Multiply by coefficients first, then do “products minus reactants” exactly.

Question 6 | Bond Enthalpy Estimate of ΔH

Student Task:

Estimate ΔH_{rxn} for: $H_2(g) + Cl_2(g) \rightarrow 2HCl(g)$ using average bond enthalpies.

Given: H–H = 436 $kJ \cdot mol^{-1}$; Cl–Cl = 243 $kJ \cdot mol^{-1}$; H–Cl = 431 $kJ \cdot mol^{-1}$.

(a) Calculate ΔH using Σ(bonds broken) − Σ(bonds formed).

(b) Explain why this value might differ from a value calculated using formation enthalpies.

Step 1 — Identification

Given bond enthalpies. Need ΔH estimate.

Relevant principle: $\Delta H \approx \Sigma E(\text{bonds broken}) - \Sigma E(\text{bonds formed})$. Breaking requires energy (+); forming releases energy (−).

Step 2 — Principle Statement

Sum energies of bonds broken in reactants and subtract energies of bonds formed in products (with coefficients).

Step 3 — Explicit Reasoning

Bonds broken: 1 H–H + 1 Cl–Cl = 436 + 243 = 679 kJ.

Bonds formed: 2 H–Cl = 2(431) = 862 kJ.

$\Delta H \approx 679 - 862 = -183$ kJ (exothermic).

(b) Why difference: bond enthalpies are average values (typically gas-phase, averaged over many molecules). Real bond energies depend on molecular environment; formation enthalpies are specific to substances and their states.

Step 4 — Validation

Sign check: forming two strong H–Cl bonds should release more energy than breaking H–H and Cl–Cl → exothermic (negative). Matches −183 kJ.

Step 5 — Final Answer

$\Delta H \approx -183$ kJ for the reaction as written; differences arise because average bond enthalpies are approximations and state/environment dependent.

Common Error

Reversing the subtraction (formed − broken) and getting the wrong sign.

Forgetting the coefficient 2 on H–Cl bonds formed.

Exam Strategy

Always write two explicit lists: 'Broken' and 'Formed' with counts. Then compute Broken − Formed. Do a sign sanity check using chemistry intuition.

Question 7 | Potential Energy Diagram Reasoning

Student Task:

A potential energy diagram shows a reaction with $\Delta H = -50$ kJ and activation energy E_a (forward) = 80 kJ.

(a) Determine E_a (reverse).

(b) Explain how a catalyst changes the diagram and what quantities remain unchanged.

(c) State one common student trap when interpreting potential energy diagrams.

Step 1 — Identification

Given: $\Delta H = -50$ kJ (exothermic), E_a forward = 80 kJ. Need E_a reverse and catalyst interpretation.

Relevant principle: $E_a(\text{reverse}) = E_a(\text{forward}) - \Delta H$ for exothermic reactions (because products are lower energy). More generally, $E_a(\text{reverse}) = E_a(\text{forward}) + |\Delta H|$ when ΔH is negative.

Step 2 — Principle Statement

Activation energies measure energy barrier height relative to the starting energy level. For exothermic reactions, the reverse barrier is higher by the magnitude of ΔH.

Step 3 — Explicit Reasoning

(a) Since $\Delta H = -50$ kJ, products are 50 kJ lower than reactants. Starting from products, you must climb the same peak, so the reverse E_a is 80 + 50 = 130 kJ.

(b) A catalyst lowers the activation energy by providing an alternate pathway with a lower peak. It does not change reactant or product energies, so ΔH remains unchanged.

(c) Trap: Students think a catalyst changes ΔH because the peak is lower. Correction: ΔH depends only on initial and final energy levels, not the pathway.

Step 4 — Validation

Check relationship: reverse barrier must be larger for exothermic because products are lower (passes).

Step 5 — Final Answer

E_a(reverse) = 130 kJ. Catalyst lowers E_a but does not change ΔH or the energy levels of reactants/products.

Common Error

Computing E_a(reverse) as 80 − 50 = 30 kJ because ΔH is negative, ignoring the physical meaning of energy levels.

Claiming catalysts change ΔH.

Exam Strategy

Sketch a quick 'energy level' picture: reactants at 0, peak at +80, products at −50. Then read reverse barrier from products to peak: 130.

Question 8 | Calorimetry Error Analysis

Student Task:

In a calorimetry experiment intended to determine ΔH_{rxn}, a student reports ΔH = −95 kJ·mol−1.

Discuss the direction of error (more negative, less negative, or unpredictable) for each issue:

(a) Heat loss to the surroundings (calorimeter not perfectly insulated).

(b) Incomplete reaction (not all limiting reagent reacts).

(c) Stirring stops intermittently causing temperature gradients.

Justify each in cause → effect form.

Step 1 — Identification

Need: direction of error for measured ΔH under specific experimental problems.

Relevant principle: The measured temperature change determines q_sur. If ΔT is underestimated, |q| is underestimated, making ΔH less negative for exothermic reactions.

Step 2 — Principle Statement

For an exothermic reaction, the true ΔH is negative. Anything that makes the measured ΔT smaller than true will make calculated ΔH less negative (closer to zero). Anything that changes the amount reacted affects the n used in ΔH = q/n and can skew results depending on calculation choices.

Step 3 — Explicit Reasoning

(a) Heat loss to surroundings: some released heat escapes without warming the measured solution. Measured ΔT is smaller → q_solution smaller → q_rxn smaller in magnitude → calculated ΔH becomes less negative (underestimates exothermicity).

(b) Incomplete reaction: If the student assumes full reaction but actually less reacts, then the heat released is smaller but they divide by the assumed larger n, making ΔH appear less negative. (If they instead measured moles actually reacted, the effect would differ; diploma typically assumes they used theoretical moles.) So: typically less negative.

(c) Poor stirring: thermometer may not read the true maximum temperature; recorded ΔT may be too low. This underestimates q and makes ΔH less negative. Random fluctuations can occur, but the common directional effect is under-recording peak temperature.

Step 4 — Validation

All three issues reduce measured temperature rise or inflate assumed reacted moles → underestimation of exothermic magnitude (less negative).

Step 5 — Final Answer

(a) Less negative. (b) Usually less negative (if assuming full reaction). (c) Usually less negative (missed peak temperature), with added random error.

Common Error

Saying heat loss makes ΔH more negative because ‘energy left the system,’ confusing the measured system (solution) with the chemical system definition used in q_rxn = −q_sur.

Treating incomplete reaction as unpredictable without stating the assumptions about n used.

Exam Strategy

Anchor on ΔT: if a problem makes ΔT smaller, it makes $|q|$ smaller. For exothermic, that pushes ΔH toward zero (less negative). State your assumptions about how n was determined.

Question 9 | Specific Heat Capacity + Proportional Reasoning

Student Task:

A 50.0 g sample of a substance is heated and its temperature rises by 12.0 °C while absorbing 2.51 kJ of heat.

(a) Determine the specific heat capacity c of the substance.

(b) If 150.0 g of the same substance absorbs 7.53 kJ, predict the temperature change assuming c is constant.

Show full unit reasoning.

Step 1 — Identification

Given: $m = 50.0$ g; $\Delta T = 12.0$ °C; $q = 2.51$ kJ. Need: c, then predict ΔT for new m and q.

Relevant principle: $q = mc\Delta T$. Convert kJ→J for standard c units ($J \cdot g^{-1} \cdot °C^{-1}$).

Step 2 — Principle Statement

Rearrange calorimetry equation: $c = q/(m\Delta T)$. For proportional scaling with constant c, $\Delta T = q/(mc)$.

Step 3 — Explicit Reasoning

(a) Convert q: 2.51 kJ = 2510 J.

$c = 2510 \text{ J} \div [(50.0 \text{ g})(12.0 \text{ °C})] = 2510 \div 600 = 4.18 \text{ J} \cdot g^{-1} \cdot °C^{-1}$.

(b) New case: $q = 7.53$ kJ = 7530 J; $m = 150.0$ g; $c = 4.18 \text{ J} \cdot g^{-1} \cdot °C^{-1}$.

$\Delta T = q/(mc) = 7530 \div [(150.0)(4.18)] = 7530 \div 627 = 12.0$ °C.

Pause and predict: triple mass and triple heat should keep ΔT the same if c is constant. Our result matches that proportional prediction.

Step 4 — Validation

Unit check: J ÷ ($g\cdot J\cdot g^{-1}\cdot °C^{-1}$) → °C (passes). Proportional reasoning check (passes).

Step 5 — Final Answer

(a) $c = 4.18\ J\cdot g^{-1}\cdot °C^{-1}$. (b) $\Delta T = 12.0\ °C$.

Common Error

Leaving q in kJ while using J-based units for c, causing c to be off by a factor of 1000.

Rearranging incorrectly (multiplying instead of dividing by $m\Delta T$).

Exam Strategy

Write q in J first unless the question explicitly asks for kJ-based c. Then use a proportional sanity check: if q and m scale together, ΔT should stay constant.

Question 10 | STS Evaluation with Quantitative Normalization

Student Task:

Two fuels are proposed for home heating: natural gas (methane) and propane.

Given standard molar enthalpies of combustion (to CO_2 and $H_2O(l)$):
• $CH_4(g)$: $\Delta H°_c = -891\ kJ\cdot mol^{-1}$
• $C_3H_8(g)$: $\Delta H°_c = -2220\ kJ\cdot mol^{-1}$

Assume prices and delivery are comparable per mole for this question.

Task: Write a structured STS evaluation that includes:
(a) quantitative comparison of energy per gram for each fuel,
(b) a discussion of CO_2 produced per kJ delivered,
(c) at least two societal/economic benefits and two environmental risks,
(d) mitigation/decision strategy with a justified conclusion.

Step 1 — Identification

Given: $\Delta H°\_c$ for methane and propane. Need energy per gram and CO_2 per kJ, then a balanced STS evaluation.

Relevant principles: energy per gram = $|\Delta H\_c|$ / molar mass; CO_2 per kJ uses stoichiometry of combustion (moles CO_2 per mole fuel) divided by energy released.

Step 2 — Principle Statement

Compare fuels fairly by normalizing to a common basis (per gram, per kJ). Then connect quantitative results to societal/environmental impacts and propose mitigation strategies.

Step 3 — Explicit Reasoning

(a) Energy per gram:

• Methane molar mass = 16.04 $g \cdot mol^{-1}$; energy per gram = 891/16.04 = 55.5 $kJ \cdot g^{-1}$.

• Propane molar mass = 44.11 $g \cdot mol^{-1}$; energy per gram = 2220/44.11 = 50.3 $kJ \cdot g^{-1}$.

So methane provides more energy per gram.

(b) CO_2 per kJ:

Combustion stoichiometry: CH_4 → 1 mol CO_2 per mol fuel; C_3H_8 → 3 mol CO_2 per mol fuel.

• Methane: 1 mol CO_2 per 891 kJ → 1/891 = 1.12×10^{-3} mol $CO_2 \cdot kJ^{-1}$.

• Propane: 3 mol CO_2 per 2220 kJ → 3/2220 = 1.35×10^{-3} mol $CO_2 \cdot kJ^{-1}$.

Thus methane produces less CO_2 per kJ delivered (lower carbon intensity).

(c) Benefits: reliable high-density energy, existing infrastructure, lower heating cost for households, grid resilience in cold climates (where electrification may strain capacity).

Risks: CO_2 emissions contribute to climate change; methane leakage (for natural gas) has high short-term warming impact; combustion produces pollutants (NO_x) affecting air quality; fossil dependence creates long-term policy/market risk.

(d) Mitigation/decision: improve furnace efficiency, leak detection and repair programs, integrate heat pumps where feasible, carbon pricing to reflect externalities, and transitional planning toward lower-carbon energy sources.

Conclusion: if choosing strictly between these two fuels under the given assumption, methane is preferable on a per-gram and per-kJ CO_2 basis, but both require mitigation and long-term transition planning.

Step 4 — Validation

Quantitative check: methane has lower molar mass and high ΔH per mole → higher $kJ \cdot g^{-1}$ (consistent). CO_2 per kJ is lower for methane due to lower carbon per energy (consistent).

Step 5 — Final Answer

Methane: 55.5 $kJ \cdot g^{-1}$ and 1.12×10^{-3} mol $CO_2 \cdot kJ^{-1}$. Propane: 50.3 $kJ \cdot g^{-1}$ and 1.35×10^{-3} mol $CO_2 \cdot kJ^{-1}$. Methane is the lower-carbon option per kJ, but both fuels carry climate and air-quality risks; mitigation and transition strategy required.

Common Error

Comparing ΔH values per mole directly without normalizing by molar mass (unfair comparison).

Using mass of CO_2 without relating it to energy delivered (missing the 'per kJ' requirement).

Exam Strategy

STS questions with data: do the math first (normalize), then interpret. Use a structure: Data → Implication → Benefit/Risk → Mitigation → Weighed conclusion.

Performance Benchmark

Performance statement:

If you can independently set up calorimetry with correct signs, scale heat using limiting reagents, manipulate Hess's Law algebraically, compute ΔH° from formation enthalpies with coefficients, estimate ΔH using bond enthalpies and justify differences, interpret potential energy diagrams, analyze experimental error directionally, and complete a quantitative STS evaluation, you are operating at a 95%+ diploma level.

Final Closure

Thermochemistry is a conservation story:

Measured temperature change → heat absorbed by surroundings → heat released/absorbed by the reaction → enthalpy per mole. Your marks come from showing each link explicitly, with units and validation.